

 CSMSS Chhatrapati Shahu Maharaj Shikshan Sanstha's CHH. SHAHU COLLEGE OF ENGINEERING Kanchanwadi, Paithan Road, Chhatrapati Sambhajinagar 431 011 	
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE	ASSIGNMENT

ACADEMIC YEAR	2025-26	COURSE TEACHER	
COURSE NAME	P&S	CLASS	S.Y.
SEM.	IV	DATE	25/04/2026

ASSIGNMENT NO. 03

Que. No.	Question Description	Marks	CO No.	BT Level														
1)	A multiple choice test consists of 8 questions with 3 answers, with no negative marking for wrong answer. Student give answer randomly, to get distinction student must solve 75% correct answers. What is the chance that student must get distinction.	5	CO3	L3														
2)	In a multiple choice examination, there are 20 questions. Each question has four alternatives answers following it and student must select the one correct answer. Four marks are given for correct answer and one mark is deducted for every wrong answer. A student must secure at least 50% of maximum possible marks to pass the examination. Suppose that a student has not studied at all so that he decides to select the answers to questions on a random basis. What is the probability that he will pass in examination?	5	CO3	L3														
3)	Fit binomial distribution to the following distribution. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>f</td><td>2</td><td>14</td><td>20</td><td>34</td><td>22</td><td>8</td></tr></table>	x	0	1	2	3	4	5	f	2	14	20	34	22	8	5	CO3	L3
x	0	1	2	3	4	5												
f	2	14	20	34	22	8												
4)	The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the probability that i) Exactly two will strike the target. ii) At least two will strike the target.	5	CO3	L3														
5)	Five coins are tossed 100 times and the following results were obtained. Fit a Binomial distribution. <table><tr><td>No. of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Frequency</td><td>10</td><td>20</td><td>30</td><td>15</td><td>15</td><td>10</td></tr></table>	No. of heads	0	1	2	3	4	5	Frequency	10	20	30	15	15	10	5	CO3	L3
No. of heads	0	1	2	3	4	5												
Frequency	10	20	30	15	15	10												
6)	In a factory of blades, there is a chance of 1/500 for any blade to be defective. Blades are supplied in a packet of 10. By using Poission distribution calculate approximate number of packets containing no defective and two defective blades in 10000 packets.	5	CO3	L3														
7)	Fit a Poisson distribution to the following <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$P(X)$</td><td>46</td><td>38</td><td>22</td><td>9</td><td>1</td></tr></table>	X	0	1	2	3	4	$P(X)$	46	38	22	9	1	5	CO3	L3		
X	0	1	2	3	4													
$P(X)$	46	38	22	9	1													
8)	If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective.	5	CO3	L3														
9)	The average number of acres burned by forest and range fires in a large New Mexico county is 4,300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal.What is the probability that between 2,500 and 4,200 acres will be burned in any given year? Given $Z(0.13) = 0.0082$, $Z(2.40) = 0.4483$	5	CO3	L3														
10)	The life of army shoes is normally distributed with mean eight months & S.D. two months. If 5000 pairs are issued, how many pairs would be expected to need replacement after 12 months? (The area under standard normal curve between o to 2 is 0.4772)	5	CO3	L3														
11)	Sack of sugar packed by an atomic loader having an average weight of 100 kg with standard deviation 0.250 kg. Assuming normal distribution find chance of sack get weighing less than 99.5 kg. (Given $A(2)=0.4772$)	5	CO3	L3														

12)	(i) In a normal distribution, 31% of the items are under 45 and 8% are over 64. find the mean and standard deviation of the distribution. (ii) What % of the items differ from the mean by number not more than 5?	5	CO3	L3
13)	Assume the mean height of soldiers to be 68.22 inches with a variance of 10.8 inches. How many soldiers in a regiment of 1,000 would you expect to be i) over six feet tall distributed. ii) below 5.5 feet? Assume heights to be normally Given: [(Area between Z=0 and Z=1.15)=0.3749, (Area between Z=0 and Z=0.67)=0.2501]	5	CO3	L3